Ex. 26. ***Find the smallest number to be added to 1000 so that 45 divides the sum exactly.***

Sol. On dividing 1000 by 45, we get 10 as remainder.

$\therefore$ Number to be added = (45 – 10) = 35.

Ex. 27. ***What least number must be subtracted from 2000 to get a number exactly divisible by 17?***

Sol. On dividing 2000 by 17, we get 11 as remainder.

$\therefore$ Required number to be subtracted = 11.

Ex. 28. ***Find the number which is nearest to 3105 and is exactly divisible by 21.***

Sol. On dividing 3105 by 21, we get 18 as remainder.

$\therefore$ Number to be added to 3105 = (21 – 18) = 3.

Hence, required number = 3105 + 3 = 3108.

Ex. 29. ***Find the smallest number of five digits which is exactly divisible by 476.***

Sol. Smallest number of 5 digits = 10000.

On dividing 10000 by 476, we get 4 as remainder.

$\therefore$ Number to be added = (476 – 4) = 472.

Hence, required number = 10472.

Ex. 30. ***Find the greatest number of five digits which is exactly divisible by 47.***

Sol. Greatest number of 5 digits is 99999.

On dividing 99999 by 47, we get 30 as remainder.

$\therefore$ Required number = (99999 – 30) = 99969.

Ex. 31. ***When a certain number is multiplied by 13, the product consists entirely of fives. Find the smallest such number.***

Sol. Clearly, we keep on dividing 55555....... by 13 till we get 0 as remainder.

$\therefore$ Required number = 42735.

```
     42735
13)555555(
   52
    35
    26
     95
     91
      45
      39
       65
       65
        ×
```

Ex. 32. ***When a certain number is multiplied by 18, the product consists entirely of 2's. What is the minimum number of 2's in the product?***

Sol. We keep on dividing 22222......... by 18 till we get 0 as remainder.

Clearly, number of 2's in the product = 9.

```
    12345679
18)222222222 (
   18
    42
    36
     62
     54
      82
      72
      102
       90
       122
       108
        142
        126
         162
         162
           ×
```

Ex. 33. ***Find the smallest number which when multiplied by 9 gives the product as 1 followed by a certain number of 7s only.***

Sol. The least number having 1 followed by 7*s*, which is divisible by 9, is 177777, as 1 + 7 + 7 + 7 + 7 + 7 = 36 (which is divisible by 9).

$\therefore$ Required number = 177777 ÷ 9 = 19753.

Ex. 34. ***What is the unit's digit in the product?***

81 × 82 × 83 × .... 89?

Sol. Required unit's digit = Unit's digit in the product 1 × 2 × 3 ×.......× 9 = 0

[$\because$ 2 × 5 = 10]

Ex. 35. ***Find the unit's digit in the product*** $(2467)^{153} \times (341)^{72}$.

Sol. Clearly, unit's digit in the given product = unit's digit in $7^{153} \times 1^{72}$.

Now, 7^4 gives unit digit 1.

$\therefore$ 7^{152} gives unit digit 1.

$\therefore$ 7^{153} gives unit digit (1 × 7) = 7. Also, 1^{72} gives unit digit 1.

Hence, unit digit in the product = (7 × 1) = 7.

Ex. 36. ***Find the unit's digit in*** $(264)^{102} + (264)^{103}$.

Sol. Required unit's digit = unit's digit in $(4)^{102} + (4)^{103}$.

Now, 4^2 gives unit digit 6.

$\therefore$ $(4)^{102}$ gives unit digit 6.

$(4)^{103}$ gives unit digit of the product (6 × 4) i.e., 4.

Hence, unit's digit in $(264)^{102} + (264)^{103}$ = unit's digit in (6 + 4) = 0.

Ex. 37. ***Find the total number of prime factors in the expression*** $(4)^{11} \times (7)^5 \times (11)^2$.

Sol. $(4)^{11} \times (7)^5 \times (11)^2 = (2 \times 2)^{11} \times (7)^5 \times (11)^2 = 2^{11} \times 2^{11} \times 7^5 \times 11^2 = 2^{22} \times 7^5 \times 11^2$.

$\therefore$ Total number of prime factors = (22 + 5 + 2) = 29.